

Ryan A. Manzuk, Ph.D.

Postdoctoral Research Associate

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EDUCATION AND PROFESSIONAL EXPERIENCE

2024 →	Princeton University , Program on Science and Global Security, Princeton, NJ <i>and</i> University of Wisconsin–Madison , Department of Nuclear Engineering, Madison, WI Postdoctoral Research Associate Project focus: remote verification of uranium mining activities and environmental impacts
2018 - 2024	Princeton University , Princeton, NJ Ph.D., Geosciences; Advisor: Adam Maloof Thesis: Reading Earth’s surface history from image archives
2016 - 2018	The Peace Corps , Guinea, West Africa Agroforestry Volunteer; Project focus: reforestation, beekeeping, and coffee cultivation
2012 - 2016	The University of Chicago , Chicago, IL B.S. with honor, Geophysical Sciences; Advisor: Mark Webster Thesis: Morphometric analysis of the Cambrian trilobite <i>Olenellus</i>

RESEARCH METHODS AND SKILLS

Field: Structure from motion 3D mapping using hand held and drone photography; grid-based mapping and sample collection in differential GPS geospatial framework; source interviews, Open source intelligence

Programming and computing: Matlab; Python; C++; optical remote sensing; hyperspectral remote sensing; deep learning; machine learning; computer vision; QGIS; G-code; Dragonfly ORS; Avizo; L^AT_EX; HTML; CSS; Adobe Suite.

Languages: Superior ability in written and spoken French, as well as superior ability in Pular (a strictly verbal West African language), based upon Peace Corps implementation of ACTFL proficiency standards.

PEER REVIEWED PUBLICATIONS (*undergraduate student I mentored)

8. MANZUK, RYAN A., AND PHILIPPE, S.
Uranium in cobalt-hydroxide exports from the Democratic Republic of the Congo.
Nature Communications (in review)
7. MANZUK, RYAN A., NADEAU, M. D., MATE, C., JACQUET, S. M., AND MALOOF, A. C.
Lithofacies vectors: Disentangling environmental variability, diagenesis, and secular change.
American Journal of Science (in review)
6. MANZUK, RYAN A., BIERI, R., ERDEN, B., SAMUELS, B. M., AND MALOOF, A. C.
Tracking agricultural land-use change with intuitive satellite methods to support stakeholder and rights-based engagement.
The Geographical Journal (in review)
5. HOWES, B. J., MEHRA, A., WILCOTS, J., GEYMAN, E. C., MANZUK, RYAN A., DEUTSCH, C. A., AND MALOOF, A. C.
The where, when, and how of ooid formation: What ooids tell us about ancient seawater chemistry.
Earth and Planetary Science Letters 637 (2024), 118697 [PDF]
4. *PANIGRAHI, I., MANZUK, RYAN A., MALOOF, A. C., AND FONG, R.
Improving data-efficient fossil segmentation via model editing.
CVPR Workshop on Learning with Limited Labeled Data for Image and Video Understanding (L3D-IVU) (2023) [PDF]

3. **MANZUK, RYAN A.**, MALOOF, A. C., KAANDORP, J. A., AND WEBSTER, M.
Branching archaeocyaths as ecosystem engineers during the Cambrian radiation.
Geobiology 21, 1 (2023), 66–85 [PDF]
2. **MANZUK, RYAN A.**, *SINGH, D., MEHRA, A., GEYMAN, E. C., EDMONSOND, S., AND MALOOF, A. C.
A high-resolution, multispectral macro-imager for geology and paleontology.
GSA Today 32, 9 (September 2022), 4–9 [PDF]
1. MEHRA, A., HOWES, B. J., **MANZUK, RYAN A.**, SPATZIER, A., SAMUELS, B. M., AND MALOOF, A. C. A novel technique for producing three-dimensional data using serial sectioning and semi-automatic image classification.
Microscopy and Microanalysis (2022), 1–16 [PDF]

INVITED TALKS

Nov. 2025	Wheaton College Symposium on Science, Justice, and Human Rights
Oct. 2025	Princeton University Science and Global Security Seminar
Feb. 2025	Harvard University Biweekly Solid Earth and Planetary Physics Seminar (BiSEPPS)
Apr. 2024	Columbia University Biology and Paleo Environment Seminars

TEACHING

Spring, 2025, 2026	SPI 593G: Satellite Imagery for Policymakers, People, and the Planet [main instructor] Introduction to remote sensing with hands-on exercises focused on use cases from humanitarian contexts to open-source intelligence Co-Instructor: Alex Glaser
Spring, 2023	GEO 370 / ENV 370: Sedimentology [assistant] Data-focused explorations of the physical and chemical processes that shape Earth’s surface through the transport and deposition of sediments. Main Instructor: Adam Maloof
Spring, 2022	GEO 376 / ENV 375 / CEE 379 / MAE 376: The Physics of Glaciers [assistant] Physics-based modeling of glaciers and data analysis. Design of tank experiments for desktop glacier simulations. Main Instructors: Ching-Yao Lai and Adam Maloof
Spring, 2021	GEO 202: Ocean, Atmosphere and Climate [assistant] Introduction to gathering and analyzing Geoscience data including satellite imagery and float-based measurements. Main Instructor: Graeme MacGilchrist
Fall, 2020	ENV 367 / GEO 367: Earth System Modeling, assessing mitigation strategies [assistant] Forcing a compact Earth system model (OSCAR) and analyzing outputs in Python Jupyter notebooks. Main Instructor: Laure Resplandy
Fall, 2019, 2020	Junior Colloquium: Princeton University Department of Geosciences [main instructor] Basic skills in data analysis and computation using Python Jupyter notebooks.

SELECTED AWARDS, HONORS, AND FELLOWSHIPS

2025-2026	Digital Science Communication Fellow, Museum of Science
2025-2026	Future Values Fellow, Princeton University
2023-2024	Myhrvold-Havranek Graduate Fellowship, Princeton University
2019-2023	Department of Geosciences Graduate Student Research Fund Princeton University [3×]
2021	Department of Geosciences Graduate Student Teaching Award, Princeton University
2021	Princeton Research Day Orange & Black Presenter Award, Princeton University
2020	Society for Sedimentary Geology Weimer Student Research Grant
2018-2019	David Elliston Guyer '67 and Walter R.F. Guyer *41 Fellowship, Princeton University
2016	Mary Jean Mulvaney Scholar Athlete Award, The University of Chicago